

2021 Ph H2 Q2

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

Two trolleys collide on a frictionless track. You are asked to apply conservation of momentum and principles of elastic collisions.

(a)

Calculate the velocity of the two trolleys after the collision

Given:

- Trolley A:

$$m_1 = 2.0 \text{ kg}, \quad v_1 = 0.50 \text{ m s}^{-1}$$

- Trolley B:

$$m_2 = 1.0 \text{ kg}, \quad v_2 = 0 \text{ m s}^{-1}$$

- After collision, the trolleys **join together**

Using **conservation of momentum**:

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_{\text{final}}$$

$$2.0 \times 0.50 + 1.0 \times 0 = 3.0 \times v_{\text{final}} \Rightarrow 1.0 = 3.0 \times v_{\text{final}} \Rightarrow v_{\text{final}} = 0.33 \text{ m s}^{-1}$$

(b)

Calculate the loss in kinetic energy during the collision

Kinetic energy **before** collision:

$$E_{\text{initial}} = \frac{1}{2} m_1 v_1^2 = \frac{1}{2} \times 2.0 \times (0.50)^2 = 0.25 \text{ J}$$

Kinetic energy **after** collision:

$$E_{\text{final}} = \frac{1}{2} (m_1 + m_2) v_{\text{final}}^2 = \frac{1}{2} \times 3.0 \times (0.33)^2 = 0.5 \times 3.0 \times 0.1089 = 0.1634 \text{ J}$$

$$\text{Loss} = E_{\text{initial}} - E_{\text{final}} = 0.25 - 0.163 = 0.087 \text{ J}$$



Revision Tips

- In **completely inelastic collisions**, objects stick together and **kinetic energy is not conserved**, but **momentum always is**.

- Use:

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_{\text{final}}$$

when objects join post-collision.

- Calculate kinetic energy **before and after** separately to check for energy loss.

- Watch for rounding — carry at least 3 sig. figs through until the final answer.